Multi-target IC v4: ic5 vs standard multi-target rho

Multi-target IC v4 results: the IC is dearer than standard multi-target rho in every cell

Protocol: `PROTOCOL.md` (frozen in 25a2540781 before data). Holdout: 33 fresh prime-order curves (group orders about 2^24, 2^28, 2^32), seeds 31 and 32, T in {1024, 4096, 16384}: **3960 runs, 0 errors, every target of every run verified**. Data: `results/holdout/rows.jsonl`, `runs.json` (compact counters in Git), `analysis.txt`, `summary.json`, `noverify.txt`. Every number here is [pcc:unverified] until the claim is signed.

Cost per target in GAE = [(c_I I + M + S)/(c_I + 3) + h (H + L)] / T; ratios are medians of 22 per-configuration ratios with bootstrap 95% intervals. icbest = the cheaper of ic5 and ic5ks per configuration and price; rhobest = the cheapest of the 15 retention-rho runs (rho2xks x 3 kappa, rhodp x 12) per configuration and price.

Result

- **icbest vs rhobest: dearer in all 27 (cell, c_I) decisions**, at every size and T. The median ratio is 1.003-1.810 over all cells, c_I and h; at I = 10 M it is 1.04-1.16 at h = 0, 1.03-1.08 at h = 0.1 and 1.20-1.76 at h = 1. ic5 alone and ic5ks alone are also dearer in all 27.
- Without the verification phase that every arm shares (`noverify.txt`, I = 10 M) the ratios are larger: 1.10-1.17 at h = 0, 1.05-1.09 at h = 0.1, 1.32-1.77 at h = 1.
- The v3 result that ic5 was cheaper than this harness's rho (0.85-0.98, PCC-20261111-dda51e) was a property of the rho reference without retention, as the v3 validator found; with the standard multi-target rules, rho wins.
- Distance to BSGS: icbest is 1.14-1.50x a paired BSGS (dearer in all 27).

Table (I = 10 M)

bits	T	icbest/rhobest h=0	h=0.1	h=1	icbest/rho2xks h=0	icbest/rhodp h=1	icbest/bsgs2 h=0	GAE/t icbest, rhobest (h=0)
24	1024	1.110 [1.102, 1.123]	1.063 [1.059, 1.079]	1.474 [1.461, 1.491]	1.110	1.474	1.353	142.3, 128.5
24	4096	1.073 [1.067, 1.075]	1.044 [1.038, 1.047]	1.322 [1.312, 1.329]	1.073	1.322	1.237	87.9, 82.0
24	16384	1.039 [1.039, 1.041]	1.028 [1.027, 1.030]	1.197 [1.196, 1.202]	1.039	1.197	1.146	61.8, 59.4
28	1024	1.148 [1.121, 1.158]	1.083 [1.056, 1.093]	1.663 [1.644, 1.680]	1.148	1.663	1.443	456.8, 401.5
28	4096	1.112 [1.106, 1.120]	1.057 [1.050, 1.066]	1.566 [1.560, 1.571]	1.112	1.566	1.371	247.2, 222.1
28	16384	1.082 [1.079, 1.084]	1.041 [1.038, 1.043]	1.446 [1.440, 1.448]	1.082	1.446	1.280	143.2, 132.4
32	1024	1.156 [1.137, 1.166]	1.083 [1.066, 1.092]	1.757 [1.741, 1.775]	1.156	1.757	1.488	1706.9, 1475.8
32	4096	1.121 [1.114, 1.125]	1.054 [1.048, 1.058]	1.681 [1.667, 1.685]	1.121	1.681	1.428	855.6, 761.2
32	16384	1.103 [1.100, 1.105]	1.043 [1.040, 1.045]	1.609 [1.604, 1.612]	1.103	1.609	1.378	448.1, 405.7

Which rho wins (per configuration, I = 10 M): at h = 0 and h = 0.1, `rho2xks` with kappa = 0.5 in all 198 configurations; at h = 1, `rhodp` in all 198 (precomputation kappa 0 or 0.25, d offset -1 to 1). kappa = 0.5 is the smallest retention-rho database in the grid, so a finer grid could only make the reference cheaper; the conclusion does not depend on it.

Why (interpretation)

ic5's table and rho's database are built at the same price per stored class (x-only shared inversions), and ic5's index-calculus phases are a few per cent of its cost (v3). What remains is coverage per field operation in the target phase: a retention rho turns every target's walk into stored, known-log points for the next target [pcc:unverified], so the effective database grows with the targets solved, while ic5's table is fixed after phase 1 (ic5ks's retained trial classes help it only 0-2%). At h = 1 the distinguished-point rho touches the hash table once per 2^d steps, against once per class for the table methods, which is why the gap is largest there. None of this uses a property of the curve.

Pre-registered predictions vs outcome

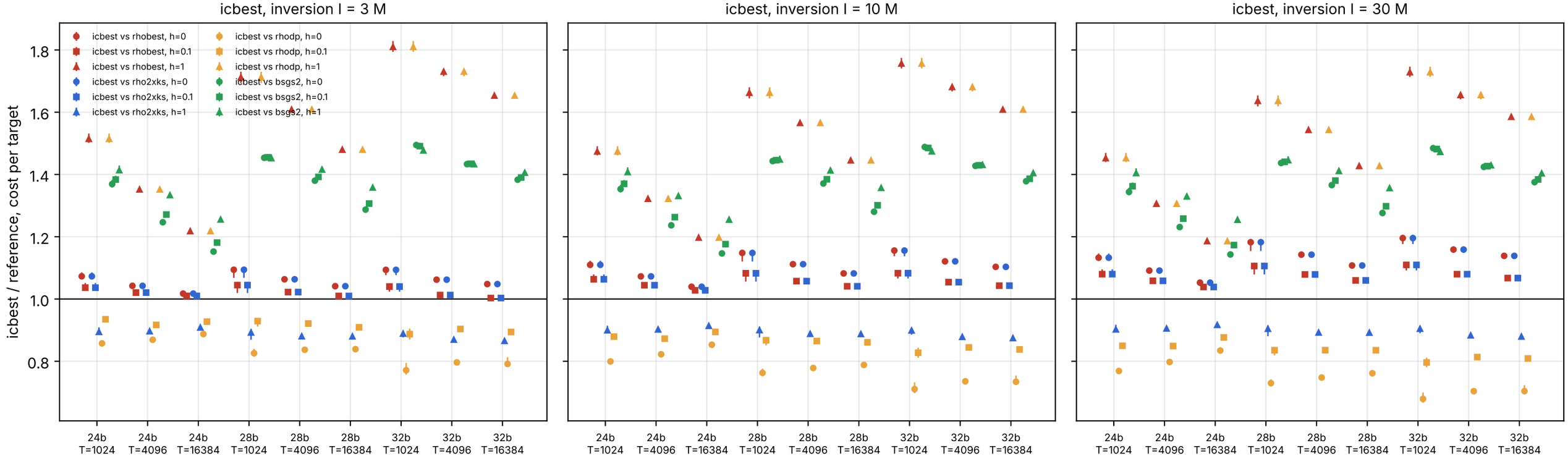
prediction	outcome
rhodp is the cheapest rho at h = 1, rho2xks at h = 0	holds (all 198 configurations each)
icbest vs rhobest: no cell cheaper at all three h	holds
icbest cheaper at h = 0 at 32 b, T = 16384	wrong: 1.103 [1.100, 1.105] (the smoke compared with rho2xks at kappa 1.0, not the grid's best)
most cells unranked, some dearer	wrong: dearer in all 27
icbest dearer than bsgs2 everywhere	holds

Answer

For the generic multi-target ECDLP on prime-order curves in this harness's counted field layer, the multi-target index calculus (ic5, the best variant built here) is not cheaper than Pollard rho [pcc:unverified] once rho uses the standard multi-target rules: it is 4% to 76% dearer per target, depending on the hash price, and 14% to 50% dearer than BSGS. Its validated improvement over the earlier IC (v3: 0.47-0.99x of ic4) stands as an engineering result for the IC baseline; the distance remaining to rho is the table above. Beating rho here would need a non-generic property of the curve (see `../mt_bsgs_gap_20261111/README.md`).

Limits

Group orders of about 2^24 to 2^32; costs counted in field operations, not timed; memory unbounded for every arm; one local host (no AWS identity, so not isolab-eks); the rho arms are this harness's (not ecbench's `rho.multitarget_pooled`), and `rhodp`'s fruitless-cycle handling (step switching plus abandoning walks over 20 * 2^d steps) is one standard choice among several; the comparison is per target in a batch of T on one curve and says nothing about single-target ECDLP or about ECC2K-130 / m = 83.


icbest, inversion l = 3 M
icbest, inversion l = 10 M
icbest, inversion l = 30 M
icbest / reference, cost per target
1.8
1.6
1.4
1.2
1.0
0.8
icbest vs rhobest, h=0
icbest vs rhobest, h=0.1
icbest vs rhobest, h=1
icbest vs rho2xks, h=0
icbest vs rho2xks, h=0.1
icbest vs rho2xks, h=1
icbest vs rhodp, h=0
icbest vs rhodp, h=0.1
icbest vs rhodp, h=1
icbest vs bsgs2, h=0
icbest vs bsgs2, h=0.1
icbest vs bsgs2, h=1
24b T=1024
24b T=4096
24b T=16384
28b T=1024
28b T=4096
28b T=16384
32b T=1024
32b T=4096
32b T=16384